

When you choose a technique to study the brain you need to think of what you want to know (spatial distribution of the processes in the brain?). EEG is cheaper and more accessible, also used in movable conditions even though there are a lot of artifacts. The main problem is that the subject is alive. Eventually you can program the use of a machine (ex. drone) relying on the eyes or muscle signal artifacts, not on the brain.

A lot of people prefer f-MRI, but the problem is what you need to get. This type of imaging technique is very fancy, if you need high temporal resolution you can't rely on it. Typically to have a more informative view you have to merge different techniques, also transcranial magnetic stimulation. There is no perfect technique, it all depends on what you want to get.

There is a position (near corona) in which there are no muscles, so no artifacts, but the speed of reaction is very low. With every heartbeat the EEG becomes very noisy, this is because the brain (the head) slightly moves inside the scanner, and the magnetic field produces a current in the EEG system.

You can get neural navigation (jump in the dark, you don't know where you are stimulating) by integrating EEG and TMS, getting the effect of the stimulation inducted, through the EEG.

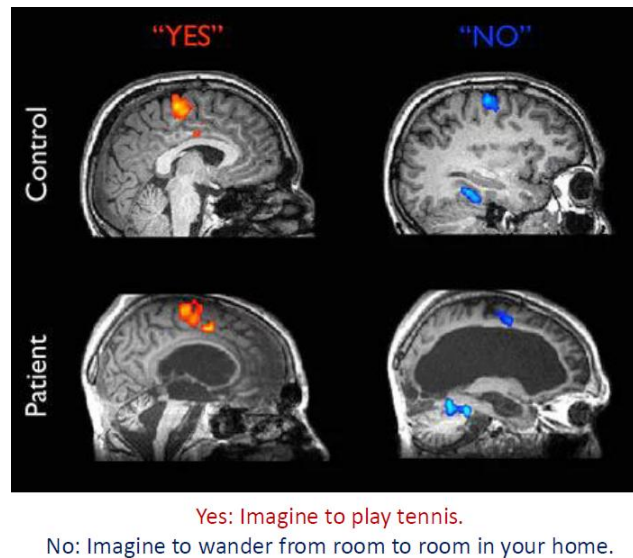
Important rule in neuroscience (Garbage In Garbage Out), meaning that if the data is collected in a bad way there is no method of analyzing that can improve the quality of the signal.

Consciousness (application of the methods described)

Consciousness is something that was not considered a science at first, because of a big debate on what parts of the brain are that makes us conscious. One theory says that any complex system can be conscious (the problem is only how conscious you are), the other says that without the frontal cortex we can't have consciousness (some animals have very small frontal cortex, or even newborns).



To say that someone is conscious you have to say what you mean by it, in this case she was conscious, having a dream. Is a person in vegetative state (coma) conscious? Seeing him you wouldn't say so, but actually he is so, and you can verify it by using the f-MRI making questions and seeing the answers.



Some studies verified that when the subject is conscious or more likely to gain consciousness and wake up, he has more complex responses, compared to when they are unconscious.

Consciousness: the ability to experience or to feel, having a sense of selfhood or soul, the fact that there is something "that it is like" to "have" or "be" it, and the executive control system of the mind, or the state or quality of awareness, or, of being aware of an external object or something within oneself. This is something that is totally inaccessible, everyone has their feelings, there is no quantitative measure to say if one person is conscious.

- **Consciousness:** intrinsic and integrated experience; it is what it "feels like" to be a system from the system's own perspective.
- **Awareness:** accessing or reporting information (*conscious access*); the cognitive ability to notice or respond to external or internal events.

Subtle but important difference. The information that something happens in the background may be processed in the brain, but I may not be aware of it. This is related to conscious access, the fact that some stimuli may be presented to the brain and be processed but I may not become conscious of this information until there is a trigger (driving a car, a lot of stuff is automatic).

We have the content (the level) and then we have attention. The process in which we direct our awareness, the attention, can be:

- **Bottom-up:** when a salient stimulus (e.g., a flash or a sound) capture our attention. This is not under voluntary control. In our brain there is a system that detects silent information and direct automatically the attention towards it. Comes from the perception.
- **Top-down:** attention willingly deployed toward a particular position in space, a particular feature, or a particular object. This is under voluntary control, when I direct attention towards something.

They are not necessarily in synergy; you may want to pay attention to something while at the same time there is noise outside and you get distracted. They influence each other. Awareness is the presence of conscious access; attention is a process that allows me to select what goes in the conscious access. Basically, the two things go together but not necessarily.

In blindsight the person has damage of the visual cortex of the brain, there is a stroke and the region of the brain is damaged (no visual consciousness), but is seen in this people that if you put them in front of a screen and you present some stimuli the person would say that he doesn't see it, but if you move the stimuli left or right and you ask him where, the person will get the right direction the majority of the time. Their ability to select the right response increases if they pay attention to the task and at the area of the screen that they don't actually see, they can guess more correctly.

Attention test → when you are focusing your attention to a very specific attention within the image you lose everything else. If I show this video while doing an f-MRI scan I can see that specific regions of the brain activate when the gorilla is present. Attention is what makes you select information.

Gist (the high-level semantic description of a scene) appears to be immune from inattentional blindness. Because gist is a property associated with the entire image, any process that locally enhances features, such as focal attention, is of limited use. When a photograph covering the entire background is briefly and unexpectedly flashed onto the screen, subjects can accurately report a summary of its content. The gist of a scene can be extracted in a period as short as 30 ms. A subject engaged in a task that requires focusing attention on the centre of the visual field, can still perceive the difference between a background scene containing an animal or a vehicle from one that does not. While the gist of a scene may be experienced vividly, it is also coarse. You can have conscious access even you don't pay attention to all the specific access.

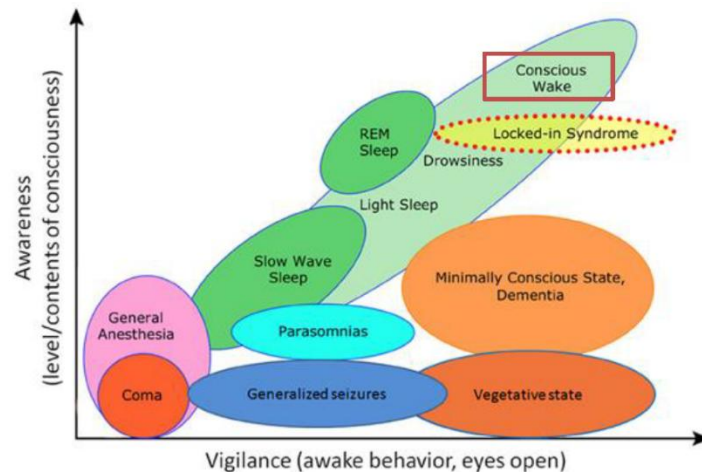
Attention it's a way to amplify information's so that you can be aware of it, really important for conscious access.

In normal conditions all of us is conscious. To study when I am or not we have to make contrast (someone that is awake vs. someone with brain damage). When I compare the two I will have a lot of differences, so it becomes really difficult to comprehend what is the thing that makes one person conscious. Each one of us has a different brain, so it would be better to compare the same brain in different conditions, but we can't, so we use other models (sleep or anaesthesia).

Under anaesthesia there is for sure an alteration in the level of consciousness and is in the same brain. Sleep is a little bit more interesting, because is a natural condition and it can be interrupted almost instantaneously with a big noise for example.

There is a condition in which consciousness is profoundly altered that is epilepsy, that is much more difficult to study (episode that last a short amount of time). There have been some studies but very few.

If you find a measure this should work for all the models. If you find that during sleep there is a brain region that goes off and it is important for consciousness, you should see the same behaviour for all the models.



This compares the content of consciousness (awareness) and the vigilance (the ability to interact with the external world). Parasomnia is a particular condition in which a person sleepwalks, it resides in that position because it is a condition where you may have a dissociated state (part of the brain awake and the other asleep), in this condition you are processing the information of the external world, but they live inside like a dream. Then lucid dream (a little above the REM sleep and close to conscious awake) is when you have a dream and you realize that you are in it, you are conscious and vigilant, aware of what happens (the frontal cortex lights up). Sleep paralysis is a condition in which you are blocked inside your body (in between REM sleep and conscious wake), in REM sleep your muscles are blocked and you become awake with a dream like experience. Brain is not all or nothing, is made of different structures the cooperate and communicate, but sometimes this communication is broken (even in physiological conditions).

A concept that is important talking about consciousness is the QUALE (sort of philosophical concept), the plural is QUALIA, and they are individual instances of subjective, conscious experience. The '*what it is like*' character of mental states, of an aspect or experience. The way it feels to have mental states such as pain, seeing red, smelling a rose, etc. We don't see all the world in the same way.



Example of different perception of something.

The reason is related to sleep regulation, our brain makes continuously assumption about what we see, we perceive a reconstruction of the world, and this is made of assumption (for example conditioned by the light). The perception of the colour of the dress may be affected by the time people go to sleep (different cardiac cycles).

Bat example:

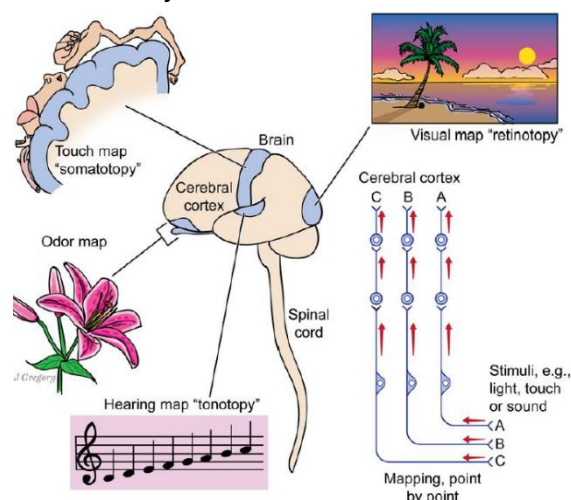
“The physical sciences can describe organisms like ourselves [...] but they cannot describe the subjective experiences of such organisms or how the world appears to their different particular points of view. There can be a purely physical description of the neurophysiological processes that give rise to an experience, and also of the physical behavior that is typically associated with it, but such a description, however complete, will leave out the subjective essence of the experience — how it is from the point of view of its subject — without which it would not be a conscious experience at all.”

Consciousness is one of the fundamental facts of human existence. Each of us is conscious. At least, one knows about his/her own consciousness directly. He/she can't be certain that other individuals are conscious. No matter how sophisticated someone's behavior is, behavior on its own is not sufficient to guarantee the presence of a conscious mind.

Turing test: proposed by Alan Turing in 1950, is a test designed to determine a machine's ability to exhibit intelligent behavior that is indistinguishable from that of a human. The test involves a human evaluator who engages in a natural language conversation with two participants: a human and a machine. The evaluator does not know which is which and must determine which participant is the human and which is the machine.

How do you determine if someone is conscious or not? You can't. This problem is known as hard problem of consciousness. The hard problem is understanding how a physical information is transformed into a subjective experience. If you believe that there is a mechanism that translate the one into the other the problem much more difficult, or you can think of it by associating this to the concept of soul.

The Neural Correlate(s) of Consciousness (NCC) is “the minimal set of neuronal events that gives rise to a specific aspect of a conscious percept”. The idea is “what do I need to get a conscious experience?”. I take the brain and start to remove pieces continuing to ask the same question until the brain doesn't sense the experience, then the last part removed was crucial for consciousness to exist and I can study it.



We have the cerebral cortex, connected to the spinal cord that is all the neurons that receive information from the periphery, we start from here. Can I cut it and continue to be conscious? Yes, but they don't have sensations and can't communicate with the external world (the person

is “locked-in”), they maintain eye movement. You can also have a complete locked-in Syndrome, where the eyes are paralyzed as well.

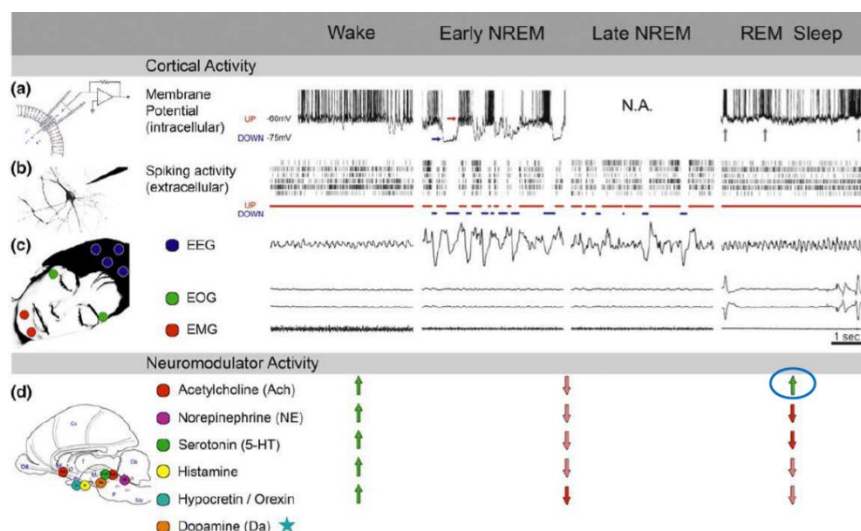
Another part of the brain that has been found to not be important for consciousness is the cerebellum, interesting because its structure (just below the brain) is crucial for acquisition of inputs, is important for learning, language. There are people that are born without it and don’t realize it’s not present, and they have some difficulties learning or with the movement.

The cerebellum contains a lot of neurons and connections (60 billion), and here they are connected in parallel, each neuron stay in a small room with some others and after the computation they send out the information that then arrives in the cerebellum, and it’s processed. So, they don’t communicate in an integrated way (with long-range communication and transversal one) like in the other parts of the brain.

Example: “A 24-year-old female patient was admitted to hospital complaining of dizziness and the inability to walk steadily for more than 20 years, and nausea and vomiting for ~1 month. She is married with a daughter, and her pregnancy and delivery were described as uneventful. [...] According to her mother, she was 4 years old before she could stand unassisted, and did not begin to walk unassisted until the age of 7, with a persistently unsteady gait. She never ran or jumped. Her speech was not intelligible until 6 years of age and she did not enter school. A neurological examination revealed she could cooperate and fully orientate. A verbal analysis test revealed her word comprehension and expression remained intact and she had no sign of aphasia, but mild to moderate signs of cerebellar dysarthria. [...] While she is able to walk unsteadily without support, her gait is moderately unsteady. The patient has evidence of tandem gait and moderately reduced gait speed. There is no focal paresis, but the muscle tone is mildly increased. Evaluation of the sensory system showed no abnormalities, no deformities of the fingers and toes were observed, and her complete blood count and urinalysis were normal.”

The cerebellum is not necessary for consciousness.

The Arousal System is a complex system, a combination of several structures. The thalamus is the structure that perceive sensor information from the periphery and send it to the cortex and it is part of the arousal system, that release neurotransmitters in the brain that keep us awake, just like noradrenaline, dopamine, histamine, glutamate, acetylcholine, serotonin, all these are attributed to different nucleus. Most of these systems are active when we are awake, and they keep up in this state.



The interesting part is that the default state of neurons is asleep (on-off state), they need the neurotransmitters to remain activate. Even without one of these systems we can remain conscious (noradrenaline-emotion danger, serotonin-emotions), these structures are needed for consciousness, if I remove these complete systems the brain is in the on-off state permanently. That is why we read articles of people waking up from long time vegetative states, because maybe the damage is not properly on the brain but in the arousal system that restart its work.

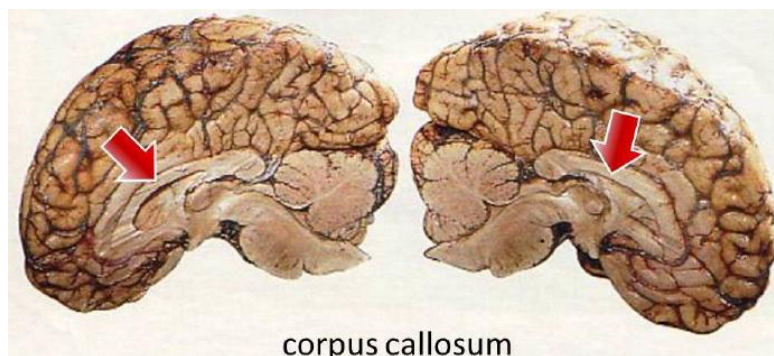
Now we are getting closer to the cortex, but the brain is composed on two hemispheres. The right hemisphere is much more literal in its recognition of images and events, and thereby often more accurate, in contrast with a left hemisphere that is constantly formulating hypotheses and trying to make sense of what happens. In addition to its superior language capabilities, the left hemisphere also retains normal problem-solving skills and reasoning abilities following callosotomy. The cognitive and linguistic abilities of the isolated left hemisphere are comparable to those of a healthy individual with an intact corpus callosum, suggesting that the left hemisphere retains a rich, human-like consciousness following split-brain surgery.

The right hemisphere's capacity for language is variable among split-brain patients and may gradually increase following surgery. The right hemisphere outperforms the left in several domains, such as perception (detection of motion, illusory contours), spatial processing (mental rotation, spatial matching, mirror image discrimination), facial recognition, and recognition memory.

What happen if you remove part or one of the two hemispheres?

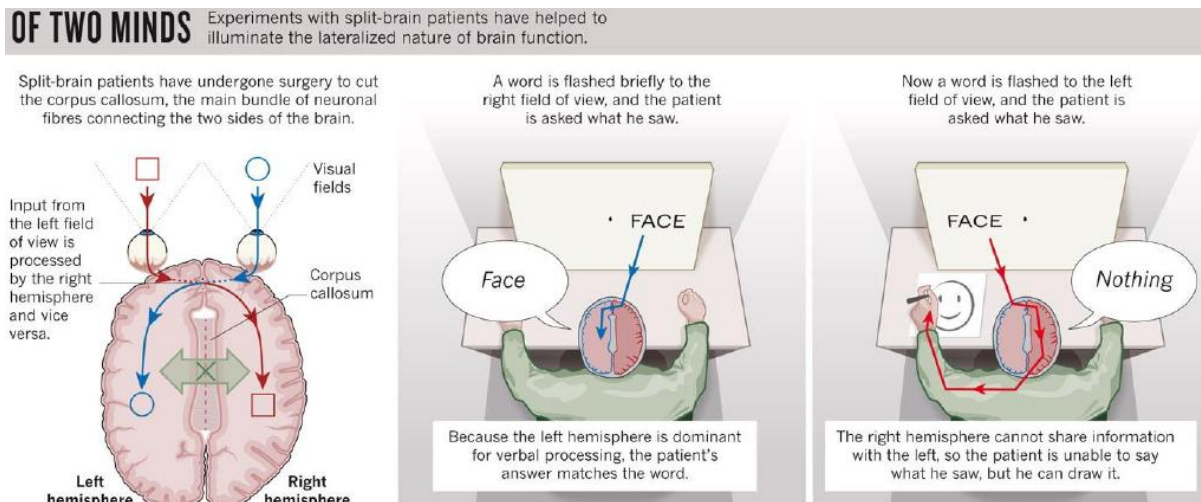
Example: a woman (neuroanatomist) got a stroke in the left hemisphere and then a complete recover. She was able to confirm all the ideas that people had about the hemisphere functions, she couldn't speak intelligently, she couldn't understand what people tell her, she was not able to think in words (internal speech). She felt a change on how she perceived the space around her, left hemisphere really important in space perception.

Example: callosotomy, the case in which the corpus callosum is cut. The arc is made of fibres that go from one hemisphere to the other, crucial for the brain to exchange information between the two hemispheres. Some people have a form of epilepsy in which seizures start from a part of the brain and propagates very fast to the other and this make the person fall of the ground (for security reasons you cut the corpus callosum). Not done anymore in almost all the world. The problem is: if one hemisphere can be conscious by itself, if I cut the connection I create two consciousness in the same person. Tricky.



Experiment: you put a person in front of a screen in which they can only see half of the screen. If I present a stimulus in the right side (perceived with the left brain), this person was able to say

that he saw a face (language is in the left brain). If you show the stimulus in the left side the person says nothing, but if you ask to draw the picture they can (right hemisphere controls drawing).

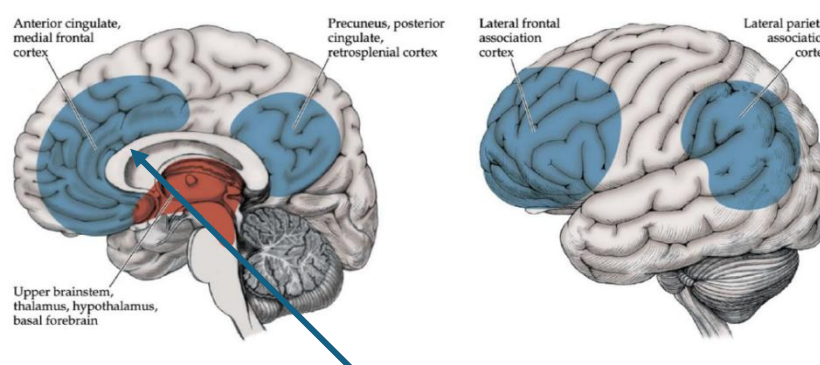


Each hemisphere of a split-brain patient cannot access the thoughts, intentions, or conscious experience of the opposite hemisphere, much as we cannot access the mental states of other individuals

Not so clearly this all is true (two consciousness in one mind), they have a really high delay. Most of them behave like child. Some of them can be smarter, maybe because the condition affected them from the birth. Now there are two theories, one says that is true (each hemisphere able to support an independent conscious self), the other one says that you don't have two separate consciousness but two separate perceptions. There was also a case in which a girl discovered to have only one hemisphere late in her life.

The brain always works to create new connections if some are lost, an example is language, even without connection with the left hemisphere the other one tries to adapt and learn how to speak (long process).

The conclusion is that the part in blue are the ones that are crucial for consciousness (present medially and laterally in the brain), the areas in red are part of the arousal system.



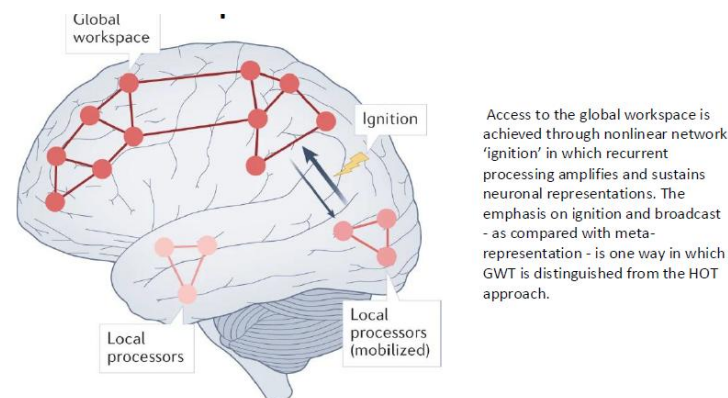
So, do we need all of the regions? The frontal cortex is crucial for decision making, interacting with the external world, producing a verbal response, to decide if I want to move or produce language (damage to this part of the brain can create a condition in which the person enters in a state of total apathy, total non-interaction, he is conscious but don't care). The role of this region is highly debated, if you pay attention, you can see that most of the top-down attention depends

on the involvement frontal cortex. If I need to pay attention to a stimulus, my frontal cortex will be activated, crucial. Is this really necessary for consciousness to happen or just necessary consciousness to be reported. This is why sleep and anaesthesia are interesting models, you don't have any voluntary control, you don't care and you forget it. Then the person only remember that they are in an experiment when they wake up (interesting to study the activation of frontal cortex while dreaming).

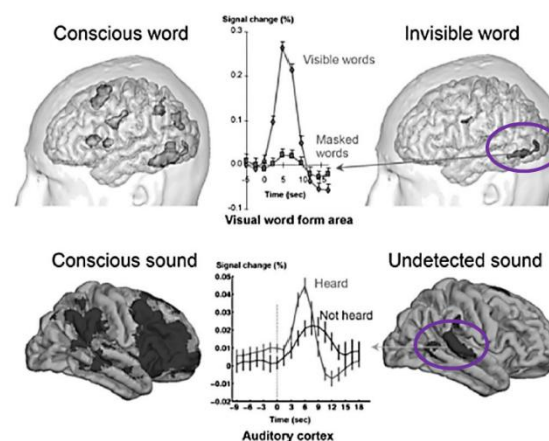
The theories of consciousness are different; many are philosophical and not scientific. If you have a theory that has no prediction and you don't test the predictions, this is not a theory.

Higher-Order theories → consciousness is mostly in the frontal cortex, able to transform experience in something that we feel. You are conscious because you reflect in you experiences.

Global workspace theories → originally proposed by Stanislas Dehaene, says that to become conscious we need also the frontal cortex. Information arrives to the posterior part of the and is processed, if this is strong enough it can reach the frontal cortex, that use an ignition, a feedback that potentiate the response (the brain creates a loop that increase the brain activation level).



There are scientific supports for this view. If I show you a subtle stimulus (embedded in noise) and it is not perceived there is an activity that is confined to the frontal cortex and does not include other regions of the brain. If you present the same stimulus but is perceived consciously you detect a much stronger activation. The ignition process is important to conscious access to happen. The information is spread through the brain and is needed for conscious access to happen. In most of the studies on these theories, you take a conscious person and present him different stimuli perceived or not and comparing them I see this type of ignition especially in the frontal cortex. These experiments can be manipulated, if in one eye the stimulus flickers then the brain will focus on the flickering stimulus, and the other one will be invisible.



Integrated information theory → everyone hates this theory because it is the most reported complete theory of consciousness (angry that their theory is not considered as much as this). There is a lot of effort by the supporters of this theory (a lot of mathematical test and prediction to try and attempt validating this theory). Any system can be conscious, depends on the connection that you have. Measure called integrated information (Φ), computed mathematically, but right now only in models, the moment you go on brain it doesn't work because it is too complicated. Φ cannot be measured on humans directly, related to entropy. The problem is that based on this idea I don't need the frontal cortex in order to be conscious, even a set of neurons that reach a certain degree of information integration can be conscious for themselves (main point of disagreement). The frontal cortex is only necessary if I need to report my experience, but then to be conscious I don't need it because the posterior regions of the brain have a sufficiently complex structure to support consciousness.

They tried to validate one of the theories as better than the others and no theory won. Only one got 2 out of 3 and the other ones 1.5 out of 3. A journalist said that Giulio Tononi's theory won (information integration) and so everyone else raised up to say that it was bullshit. The effect in which scientists are against each other is not good for research.

Information is high where there is integration and communication.

Lesson 22/10/2025 (Learning and memory)

What happens in the brain when we learn and acquire information?

Learning is the process that leads to the acquisition of new knowledge or skills, memory is the product of it (acquired knowledge and skills). Learning leads to some kind of changes in the brain that allows the brain to store the information acquired. Typically we go through 3 steps while learning:

- Encoding, that is the storage of information.
- Consolidation, most of it happens during sleep.
- Retrieval, the one in which you access the memory, and you use it in behaviour.

To acquire information, we need to rely in our senses, not only the main 5 but also through the interoceptive information processing, that acquires information from the state of the body. We learn to define emotional states based on what we feel in specific conditions (sweating, acceleration in the heartbeat, etc.).

Once that the information reaches the brain there is a sort of sensory memory, which is the most basic but also fleeting (few ms), that is the time of processing of the information that we acquire. The information stored somewhere in the brain, and this is called short-term memory, the other name is working memory. From this type of memory you can retrieve and use information, or send it in long time memory through consolidation, that continues for days, months or even years, at least few hours.

There can be also some loss of information, that can happen at all the steps, and it also may happen in all the moments in which we retrieve information from memory, and they become less accessible each time. If you learn something for an exam you see that memory weakens over time, in a point that you don't remember almost anything.

The terms short-term memory and working memory are strongly related (almost synonymous), even though the second one is more recent, introduced after realizing that when something is